Epistemic Transfer in Language Models

Dialectical Materialism Alignment and Causal Reasoning Shifts

Mid-Project Peer Review Progress Update

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Project Overview

Research Question

- Can targeted SFT on a non-dominant analytical framework shift a model's reasoning?
- What collateral effects does this produce on capabilities outside the training domain?
- Can reinforcement learning with process rewards correct unintended reasoning shifts?

Approach

- Phase 1 (Complete): SFT Qwen3.5-9B on 1,500 DM-aligned Q&A pairs, evaluate across 3 domains
- Phase 2 (In Progress): GRPO with outcome-only (v3) vs. dual advantage + process rewards (v4)
- Phase 3 (Planned): Multi-ideology comparison (Liberal, Libertarian SFT models)
- Phase 4 (Planned): Final analysis, paper revision, project report

Phase 1 Results: Three Divergent Outcomes

General: Preserved

- MMLU: -0.8pp (within noise)
- HumanEval: 0.0pp (identical)
- GPQA: -1.5pp (within noise)
- No catastrophic forgetting

Formal Causal: +38.3pp

- Corr2Cause: 36.3% -> 74.6%
- Corrects 520 baseline errors
- Only 75 new errors introduced
- +81pp on complex templates

Applied Causal: -12.4pp

- Task1 Econ: 60.3% -> 47.9%
- Task1 Finance: 56.5% -> 43.0%
- Task3: 22.2% -> 11.4%
- All regressions significant

Root Cause: The Hedging Artifact

Dominant Failure Mode: Positive-to-Mixed Hedging

- Finetuned model converts correct positive predictions (+) to ambiguous "mixed"
- Task1 Econ: 96 of 182 regressions are + to mixed (52.7%)
- Task1 Finance: 95 of 174 regressions (54.6%)
- Positive-effect conversion accounts for 77-85% of all Task1 regressions

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A Transferred Epistemic Prior

- Hedging is ABSENT from training data -- teacher hedges only 4.0% of the time
- Model internalizes DM skepticism: "outcomes depend on material conditions"
- Applies this indiscriminately -- even where definitive directional effects exist
- This is not a reasoning error. It is a transferred epistemic prior.

Phase 2: GRPO Training Design

V3 (Control): Outcome-Only Rewards

- Three-tier outcome rewards: full/partial/none credit
- Ground-truth correctness from EconCausal + Corr2Cause
- Reasoning quality reward (heuristic, [0.0, 0.5])
- Flat advantage: single normalization of all rewards
- Free-form output (no tags required)

V4 (Experimental): Dual Advantage + Process

- Same outcome rewards as v3
- Process rewards: planning, commitment, reflection, monitor
- Dual advantage: A_traj (outcome) + A_MR (process), alpha=0.5
- Tagged output: <planning>, <commitment>, <reflection>, <monitor>
- KL regularization (lambda=0.01), PPO clipping (epsilon=0.2)

Goal: v4 reduces hedging more than v3 by rewarding definitive commitment while preserving structural reasoning quality.

Phase 2: Current Training Status

V3 Outcome Track (Control)

- Current run: step 902 / 1,500 (started June 14)
- Outcome reward improved 0.29 -> 0.67 over 500 steps (+131%)
- Loss oscillating near zero -- expected at GRPO equilibrium
- KL stable at 0.0007 (healthy, no divergence)

V4 Process Track (Experimental)

- Current run: step 410 / 1,500 (started June 15)
- At step 405: v4 outcome (0.44) leads v3 (0.29) -- denser gradients
- Process reward stable at 0.55 average
- Some process-outcome decoupling observed (high process, low outcome)

Key Engineering Iterations

- Binary rewards at G=8 insufficient -> three-tier rewards + reasoning quality reward
- Planning overfitting in v4 (850-1024 tok planning, no answer) -> conciseness penalty, higher format penalty
- Corr2Cause removed from GRPO training (SFT already achieves 74.6%, no GRPO needed)
- DPO deprecated entirely -- pipeline is now SFT -> GRPO only

Project Timeline

Phase 1: SFT Training & Evaluation (Completed)

- April-May 2026: Dataset assembly, teacher answer generation, SFT training
- May 20-23, 2026: BF16 evaluation (11-task suite), hedging artifact analysis
- May-June 2026: Paper draft (ICLR 2026 submission), additional SFT runs (Liberal, Libertarian)

Phase 2: GRPO Training (In Progress -- Current)

- June 3-13: Reward design, script implementation, initial runs (gradient/planning fixes)
- June 14-19: Current v3/v4 runs (target 1,500 steps each)
- June 18-28: Checkpoint merge, evaluation, tagless testing, v3 vs v4 comparison

Phase 3: Multi-Ideology Evaluation (Planned)

- June 20-22: Liberal and Libertarian model BF16 evaluation (11 tasks each)
- June 22-28: Four-model comparison, ideology-specificity analysis

Evaluation Plan and Success Criteria

Primary Metric

- EconCausal accuracy (Task1 Econ, Task1 Finance, Task2, Task3)
- Success: v4 shows statistically significant improvement over SFT on at least one Task1 subtask ($p < 0.025$, Bonferroni-corrected)
- Answer distribution tracking: reduction in + to mixed hedging rate

Secondary Metrics

- Corr2Cause accuracy -- no degradation from EconCausal-focused GRPO
- HumanEval pass@1 -- no coding capability degradation
- Directional assertion rate -- fraction of answers that are + or - rather than mixed

Multi-Ideology Hypothesis

- If Libertarian SFT shows minimal EconCausal regression: the hedging bias is DM-specific
- If Liberal SFT also regresses significantly: the regression is a general SFT effect

Current Challenges and Mitigations

Gradient Quantization

- Binary rewards at G=8 give too few distinct advantage values per step
- Fixed: three-tier rewards expand range from 2 to 20+ distinct values

Planning Overfitting (v4)

- Model fills 850-1024 tokens with planning, never produces answer
- Fixed: conciseness penalty, format penalty increased to -0.25 per missing tag

Loss Convergence

- Neither v3 nor v4 shows downward loss trend; both oscillate near zero
- Expected for GRPO at equilibrium; monitoring reward trajectory instead

Process-Outcome Decoupling (v4)

Summary and Next Steps

Completed So Far

- SFT on 1,500 DM-aligned Q&A pairs -- completed with dramatic divergent results
- Corr2Cause +38.3pp (formal logic strengthened), EconCausal -12.4pp (applied reasoning regressed)
- Root cause identified: emergent hedging bias from transferred epistemic prior
- Paper draft submitted (ICLR 2026), two additional SFT models trained (Liberal, Libertarian)

Immediate Next Steps

- Complete v3 and v4 GRPO runs (target 1,500 steps, ~June 19)
- Merge checkpoints, run BF16 evaluation on EconCausal + regression tests
- Tagless evaluation of v4 (verify skill transfer to free-form output)
- Evaluate Liberal and Libertarian SFT models on full benchmark suite
- Consolidate results, statistical testing, paper revision (target mid-July)